PERFORMANCE TEST RESULTS

ConstructPro ERP

1. Executive Summary

All 25 tested API routes satisfied their success and latency criteria in the 1-session, 10-session, and 50-session bounded workloads and in the 50-worker sustained diagnostic. The 50-session bounded workload completed 1,250 requests with no errors in 43.211 seconds (28.46 requests/second). The sustained diagnostic completed 2,458 requests with no errors in 62.386 seconds (39.40 requests/second).

The frontend browser diagnostic successfully authenticated through the application interface and examined eight dashboard pages. Every tested page returned HTTP 200, and all 69 recorded API responses completed without reported API errors. Frontend measurements are development-mode observations and are not production performance certifications.

2. Test Environment and Service Readiness

Item	Configuration
API gateway	Port 4000; gateway entry point for performance requests
Frontend	Port 3100; NEXT_PUBLIC_API_URL configured in the test process to point to the API gateway
Runtime	v24.12.0; Windows; AMD Ryzen 7 8845HS w/ Radeon 780M Graphics; 16 logical CPUs; 15.3 GiB RAM
Database	Existing production DATABASE_URL from backend/.env; no migrations or database resets
Backend mode	Fresh successful TypeScript compilation; 14 independent Node.js processes with NODE_ENV=production
Frontend mode	Webpack development server; production build unavailable as detailed below
Downstream routing	Gateway downstream service URLs configured using child-process environment overrides
Calendar qualification	Ephemeral runtime-only encryption key for synthetic-account status reads; no existing Google tokens decrypted or changed.

Item	Configuration
Identities	One synthetic ADMIN plus one synthetic CLIENT_PORTAL_USER per API experiment; random passwords, bcrypt cost 12; identities shared by simulated sessions
Fixtures	Synthetic lead, customer, planning project, draft quotation and draft invoice; zero monetary values; existing business records not reassigned
Cleanup	Bounded: completed; sustained: completed; browser: completed; service ports still listening after shutdown: 0

3. Workload Design and Acceptance Criteria

A GET-only runner exercised 25 routes. Each bounded session traversed the routes sequentially, with no ramp-up, retry, or think time. Authentication completed before measurement. At most 50 requests were in flight, meaning that 50 sessions did not send 50 simultaneous requests to every route. Each request had a 10-second timeout and a 2 MiB response-body limit. Measured latency included full response-body consumption, application processing, and database operations.

A route passed when every sample returned a 2xx response and its unrounded 95th-percentile latency (p95) was at most 2,000 ms, except dashboard-summary, whose threshold was 3,000 ms. Percentiles used nearest-rank computation, including failed-request durations. A single observation per route at one session is not a statistically robust tail-latency estimate.

The 50-worker sustained diagnostic scheduled round-robin GET requests for 60 seconds with closed-loop workers and then drained outstanding requests. The total elapsed time could therefore exceed 60 seconds. This was a short sustained diagnostic, not a long-duration soak or a maximum-throughput capacity test.

4. Experiment Summary

Experiment	Start time	Duration (s)	Requests	Errors
1-session bounded wave	22:51:43	28.938	25	0
10-session bounded wave	22:52:12	23.804	250	0
50-session bounded wave	22:52:36	43.211	1250	0
50-worker sustained diagnostic	22:53:24	62.386	2458	0

Experiment	Error rate	Total req/s	Successful req/s	Routes passed
1-session bounded wave	0.00%	0.86	0.86	25/25
10-session bounded wave	0.00%	10.50	10.50	25/25
50-session bounded wave	0.00%	28.46	28.46	25/25
50-worker sustained diagnostic	0.00%	39.40	39.40	25/25

HTTP status outcomes

Experiment	HTTP status counts
1 session	200: 25
10 sessions	200: 250
50 sessions	200: 1250; 500: 0
Sustained 50 workers	200: 2458

5. Bounded-Workload Results by Route

The 1-session wave recorded one latency sample per route; the 10-session wave recorded ten; and the 50-session wave recorded fifty. All latencies and thresholds in the tables below are in milliseconds.

5.1. One-session and ten-session measurements

Route	Limit	1-session	10-session p95	10 errors
current-user	2000	597	916	0/10
users	2000	1787	895	0/10
clients	2000	860	763	0/10
leads	2000	1844	1566	0/10
quotations	2000	1404	1211	0/10
projects	2000	1786	1513	0/10
tasks	2000	1447	1329	0/10
invoices	2000	1511	1470	0/10

Route	Limit	1-session	10-session p95	10 errors
calendar-status	2000	1122	1247	0/10
document-categories	2000	1085	967	0/10
dashboard-summary	3000	886	1418	0/10
revenue-kpis	2000	353	894	0/10
project-detail	2000	1012	849	0/10
project-milestones	2000	684	745	0/10
project-tasks	2000	1099	1142	0/10
expense-summary	2000	742	738	0/10
project-documents	2000	1418	1368	0/10
project-risk	2000	1484	1148	0/10
invoice-detail	2000	1321	701	0/10
invoice-payments	2000	1246	1125	0/10
client-portal-projects	2000	1860	1378	0/10
client-portal-project-detail	2000	600	689	0/10
client-portal-milestones	2000	1146	803	0/10
client-portal-documents	2000	1039	1107	0/10
client-portal-quotations	2000	604	699	0/10

5.2. Fifty-session measurements

Route	p50	p95	p99	Errors	Result
current-user	1332	1626	1893	0/50	PASS
users	1417	1506	1537	0/50	PASS
clients	815	1184	1207	0/50	PASS
leads	819	1540	1693	0/50	PASS
quotations	1007	1394	1415	0/50	PASS

Route	p50	p95	p99	Errors	Result
projects	1581	1625	1818	0/50	PASS
tasks	1197	1511	1541	0/50	PASS
invoices	1610	1984	2034	0/50	PASS
calendar-status	1545	1908	1934	0/50	PASS
document-categories	1174	1322	1361	0/50	PASS
dashboard-summary	1297	1540	1631	0/50	PASS
revenue-kpis	1221	1507	1592	0/50	PASS
project-detail	1366	1782	1936	0/50	PASS
project-milestones	1333	1629	1852	0/50	PASS
project-tasks	1195	1552	1613	0/50	PASS
expense-summary	1236	1632	1904	0/50	PASS
project-documents	1598	1950	2040	0/50	PASS
project-risk	833	1035	1135	0/50	PASS
invoice-detail	870	1033	1052	0/50	PASS
invoice-payments	1136	1338	1405	0/50	PASS
client-portal-projects	1626	1704	1836	0/50	PASS
client-portal-project-detail	1440	1702	1932	0/50	PASS
client-portal-milestones	1203	1531	1883	0/50	PASS
client-portal-documents	1695	1837	1964	0/50	PASS
client-portal-quotations	1587	1641	1777	0/50	PASS

6. Sustained-Workload Results by Route

Route	Samples	p50	p95	p99	Errors	Result
current-user	99	1427	1550	1634	0	PASS
users	99	956	1172	1701	0	PASS

Route	Samples	p50	p95	p99	Errors	Result
clients	99	843	962	1213	0	PASS
leads	99	1673	1838	2479	0	PASS
quotations	99	965	1064	2183	0	PASS
projects	99	1614	1821	2391	0	PASS
tasks	99	1393	1535	2284	0	PASS
invoices	99	1566	1902	2452	0	PASS
calendar-status	98	1518	1666	2014	0	PASS
document-categories	98	1042	1485	1967	0	PASS
dashboard-summary	98	821	922	1494	0	PASS
revenue-kpis	98	816	930	1498	0	PASS
project-detail	98	1183	1293	2152	0	PASS
project-milestones	98	1198	1328	2124	0	PASS
project-tasks	98	1534	1725	2367	0	PASS
expense-summary	98	1192	1297	2212	0	PASS
project-documents	98	1806	1951	2732	0	PASS
project-risk	98	958	1068	1966	0	PASS
invoice-detail	98	959	1225	2238	0	PASS
invoice-payments	98	1208	1339	2147	0	PASS
client-portal-projects	98	1076	1199	1745	0	PASS
client-portal-project-detail	98	1065	1197	1774	0	PASS
client-portal-milestones	98	1192	1381	1911	0	PASS
client-portal-documents	98	1317	1556	2061	0	PASS
client-portal-quotations	98	1082	1268	1811	0	PASS

7. Frontend Browser Diagnostics

Browser: Microsoft Edge 155.0.4283.62. The frontend ran in Webpack development mode against the compiled backend and production database. The session authenticated through the ordinary user interface and shared Axios client. One full navigation was measured for each of the eight pages.

TTFB denotes time to first byte; DOM is DOMContentLoaded; FCP is first contentful paint; LCP is largest contentful paint; CLS is cumulative layout shift; and Settled denotes the network-idle settling measurement. Timings are in milliseconds. These are single-browser diagnostic observations affected by development compilation and application warmness.

7.1. Navigation and rendering timings

Page	HTTP	TTFB	DOM	FCP	LCP	Settled
/dashboard	200	5432	5470	5672	8888	10405
/dashboard/users	200	1450	1540	1556	4528	5015
/dashboard/leads	200	2145	2229	2240	8180	9411
/dashboard/quotations	200	1354	1391	1416	4416	6711
/dashboard/projects	200	2498	2538	2568	4672	9295
/dashboard/finance	200	1406	1460	1480	5256	5703
/dashboard/analytics	200	1283	1318	1352	4816	5251
/dashboard/documents	200	1201	1289	1308	3280	3753

7.2. Layout stability and API outcomes

Page	CLS observed	API responses	API errors
/dashboard	0.07192	18	0
/dashboard/users	0.28898	4	0
/dashboard/leads	0.01136	7	0
/dashboard/quotations	0.00000	5	0
/dashboard/projects	0.00938	18	0
/dashboard/finance	0.00212	4	0

Page	CLS observed	API responses	API errors
/dashboard/analytics	0.00000	11	0
/dashboard/documents	0.00000	2	0

8. Interpretation

At the highest tested bounded load, the API returned successful 2xx responses for all 1,250 requests, and every route met its p95 latency threshold. The sustained diagnostic also passed all 25 routes without request errors. These results provide evidence of stable API responses for the tested GET workload under the specified concurrency, test data, and runtime conditions.

The gateway authentication guard calls /auth/me for protected requests, resulting in database reads. Business handlers may perform additional queries, and client-portal routes apply ownership checks. These operations can add database round trips. However, database query timing and query execution plans were not collected, so no specific database bottleneck can be established from these results